A SUFFICIENT CONDITION FOR THE CANONICAL MAP INTO THE CANDIDATE K-INJECTIVE COMPLEX TO BE A QUASI-ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument: it names the general criterion reducing the assertion to the derived limit statement and the companion criterion which supplies it. Both sentences are kept verbatim inside the counted proof span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The standing set-up: the construction of the candidate K-injective complex and the canonical map into it. Same chapter, the set-up of the section delivered verbatim as a context span in this card (source lines 5548-5584), including the displayed equation that the counted statement refers to. Supplies the map whose being a quasi-isomorphism is asserted.
- The companion criterion recovering an object as the derived limit of its truncations, and the general K-injectivity criterion. Same chapter, the companion results on the derived limit of the truncations and on K-injective complexes. Supplies the two results to which the proof reduces the assertion.

Source reference. Original lemma 08U3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5417–5417. Navigation pointer invoked at proof line 5598; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070F: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5554–5554. Navigation pointer invoked at proof

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line 5554; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5575–5575. Navigation pointer invoked at proof line 5575; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070L: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5576–5576. Navigation pointer invoked at proof line 5576; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 070M: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5604–5604. Navigation pointer invoked at proof line 5604; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6P: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5440–5440. Navigation pointer invoked at proof line 5607; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}^\bullet$ be a complex of $\mathcal{O}$-modules. The category $Mod(\mathcal{O})$ has enough injectives, hence we can use Derived Categories, Lemma 070F produce a diagram

$$\begin{array}{ccccc} \ldots & \longrightarrow & \tau_{\geq -2}\mathcal{F}^\bullet & \longrightarrow & \tau_{\geq -1}\mathcal{F}^\bullet \\ & & \downarrow & & \downarrow \\ \ldots & \longrightarrow & \mathcal{I}_2^\bullet & \longrightarrow & \mathcal{I}_1^\bullet \end{array}$$

in the category of complexes of $\mathcal{O}$-modules such that

(1) the vertical arrows are quasi-isomorphisms,
(2) $\mathcal{I}_n^\bullet$ is a bounded below complex of injectives,
(3) the arrows $\mathcal{I}_{n+1}^\bullet \to \mathcal{I}_n^\bullet$ are termwise split surjections.

The category of $\mathcal{O}$-modules has limits (they are computed on the level of presheaves), hence we can form the termwise limit $\mathcal{I}^\bullet = \lim_n \mathcal{I}_n^\bullet$. By Derived Categories, Lemmas 070J and 070L this is a K-injective complex. In general the canonical map

$$\mathcal{F}^\bullet \to \mathcal{I}^\bullet \tag{0.0.1}$$

may not be a quasi-isomorphism. In the following lemma we describe some conditions under which it is.

Lemma 0.1. *In the situation described above. Denote $\mathcal{H}^m = H^m(\mathcal{F}^\bullet)$ the mth cohomology sheaf. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset. Let $d \in \mathbf{N}$. Assume*

(1) *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$,*
(2) *for every $U \in \mathcal{B}$ we have $H^p(U, \mathcal{H}^q) = 0$ for $p > d$ and $q < 0$*[1].

Then (0.0.1) is a quasi-isomorphism.

[1]It suffices if $\forall m$, $\exists p(m)$, $H^p(U.\mathcal{H}^{m-p}) = 0$ for $p > p(m)$, see Lemma 08U3.

Proof. By Derived Categories, Lemma 070M it suffices to show that the map $\mathcal{F}^\bullet \to R\lim \tau_{\geq -n}\mathcal{F}^\bullet$ is an isomorphism. This follows from Lemma 0D6P. □